

Penrose diagrams

Definition of a Penrose diagram

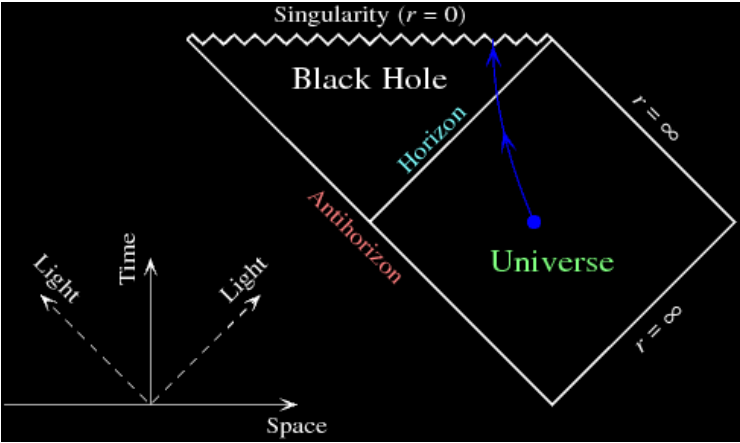
A Penrose diagram is a kind of spacetime diagram arranged to make clear the complete causal structure of any given geometry. They are an indispensable map for navigating inside a black hole. Roger Penrose, who invented this kind of diagram in the early 1950s, himself calls them conformal diagrams.

In a Penrose diagram:

- Light rays move at 45° from the upward vertical;
- Points at infinity (at infinite distance, or in the infinite past or future) are contained in the diagram.

A detailed exposition, complete with animated spacetime diagrams, on how the Penrose diagram of the Schwarzschild geometry is constructed can be found at [More about the Schwarzschild Geometry](#) on the [Falling into a Black Hole](#) website.

Penrose diagram of a Schwarzschild black hole



[Printable version \(pdf\)](#)

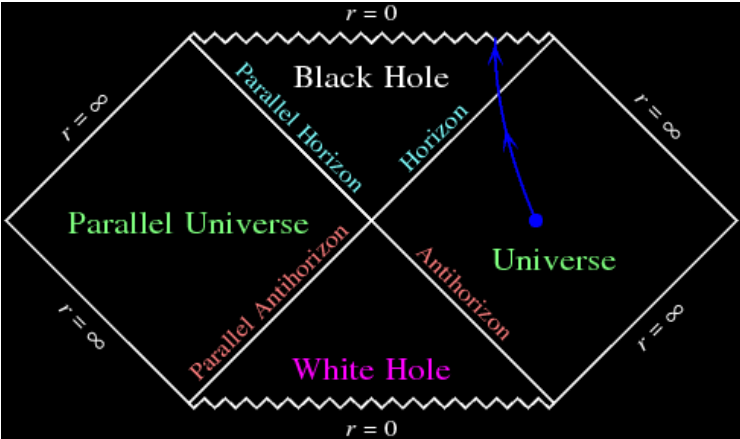
The movie of the journey into a [Schwarzschild](#) black hole showed that the horizon appeared to split into two when you fell through it. The Penrose diagram of the Schwarzschild geometry clarifies this unexpected behavior.

The Penrose diagram shows that the horizon is really two distinct entities, the Horizon, and the Antihorizon. The Horizon is sometimes called the true horizon. It's the horizon you actually fall through if you fall into a black hole. The Antihorizon might reasonably be called the illusory horizon. In a real black hole formed from the collapse of the core of a star, the illusory horizon is replaced by an exponentially redshifting image of the collapsing star. As the collapsing star settles towards its final [no-hair](#) state, its appearance tends to that of a no-hair black hole.

This [animated gif \(12K\)](#) version of the Penrose diagram illustrates light rays that start from the Antihorizon, or from the Horizon, and hit the observer. The diagram shows that when you look at a black hole from the outside, you are looking at its Antihorizon, or illusory horizon. When you fall through the horizon, you fall not through the

Antihorizon, but rather through the Horizon, or true horizon. The Horizon becomes visible only after you have fallen through it. The Antihorizon continues to remain ahead of you, and you never fall through it.

Penrose diagram of the complete, analytically extended Schwarzschild geometry



[Printable version \(pdf\)](#)

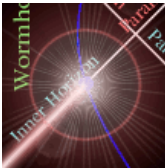
What is beyond the Antihorizon of a black hole? In a real black hole, the other side of the Antihorizon, or illusory horizon, is the star that collapsed, redshifted to blackness.

However, the Schwarzschild geometry has a simple mathematical form, and that form can be extended analytically. The mathematical extension consists of a second copy of the Schwarzschild geometry, reversed in time, glued along the Antihorizon. The complete analytic extension of the Schwarzschild geometry contains not only a Universe and a Black Hole, but also a Parallel Universe and a White Hole. This is simply a mathematical construction, with no basis in reality. Still, it is cute that even the simplest kind of black hole, a Schwarzschild black hole, harbors alien mathematical passageways.

Penrose diagram of a Reissner-Nordström black hole

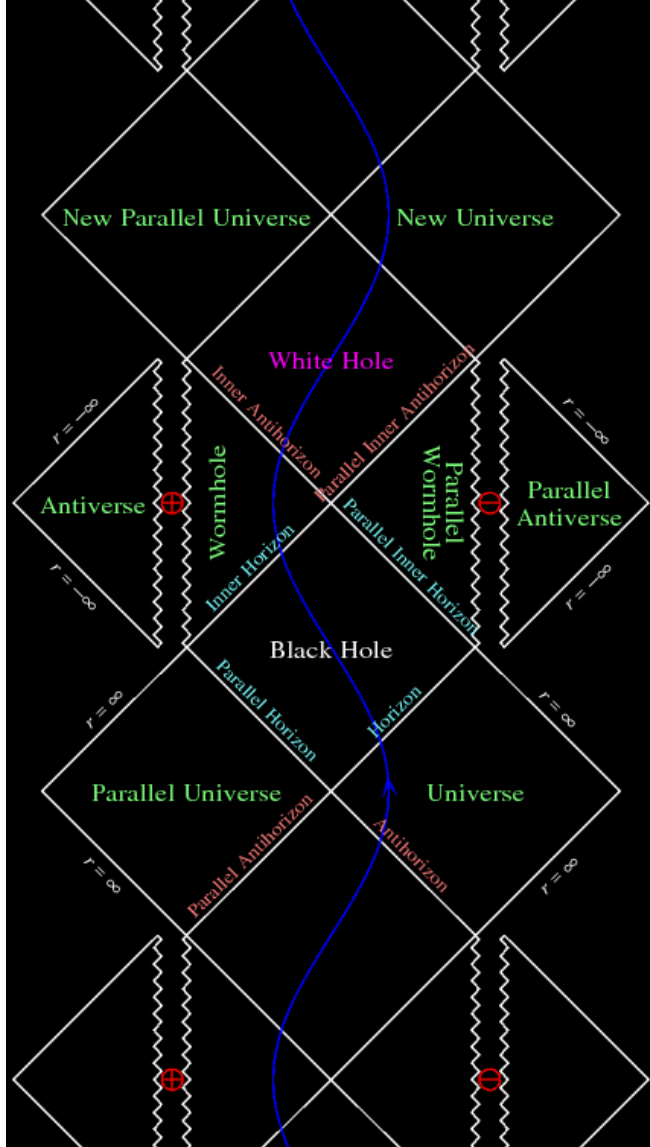
The Penrose diagram helps to make sense of the strange experience on your journey into a [Reissner-Nordström](#) black hole.

This [animated gif \(90K\)](#) version of the Penrose diagram illustrates how an infaller sees infinitely bright, infinitely blueshifted



bursts of light:

- On passing inward through the inner horizon, the infaller sees the infinite past of the Universe reflected in the gravitationally repulsive singularity;
- On passing back outward through the inner horizon, the infaller sees the infinite future of the Universe;
- On passing outward through the outer horizon of the white hole, the infaller sees the infinite past of the New Universe.



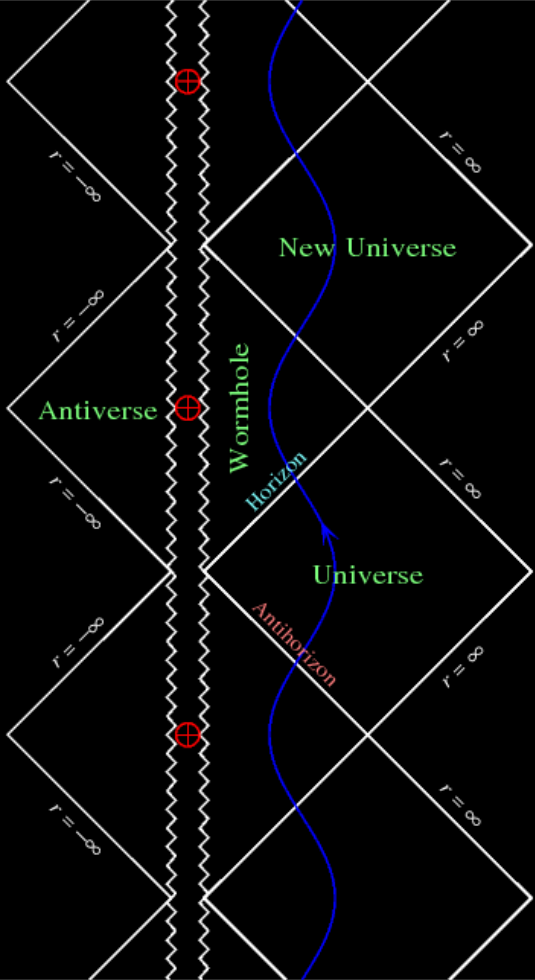
Penrose diagram of an extremal Reissner-Nordström black hole

The inner and outer horizons of a Reissner-Nordström (charged) black hole occur at radii $r_{\pm} = M \pm (M^2 - Q^2)^{1/2}$. Horizons exist only if the black hole's charge Q is less than or equal to its mass M , in natural units. The extreme case where the charge equals the mass is called an extremal black hole. For an extremal black hole, the inner and outer horizons coincide.

As illustrated by its Penrose diagram, the topology of an extremal Reissner-Nordström black hole differs from the standard Reissner-Nordström topology. For this reason and others, it is thought that an extremal Reissner-Nordström black hole cannot be constructed, even in principle, by adding charge to a standard Reissner-Nordström black hole. Indeed, adding charge to a near-extremal black hole takes energy, which increases the mass of the black hole, keeping it sub-extremal.

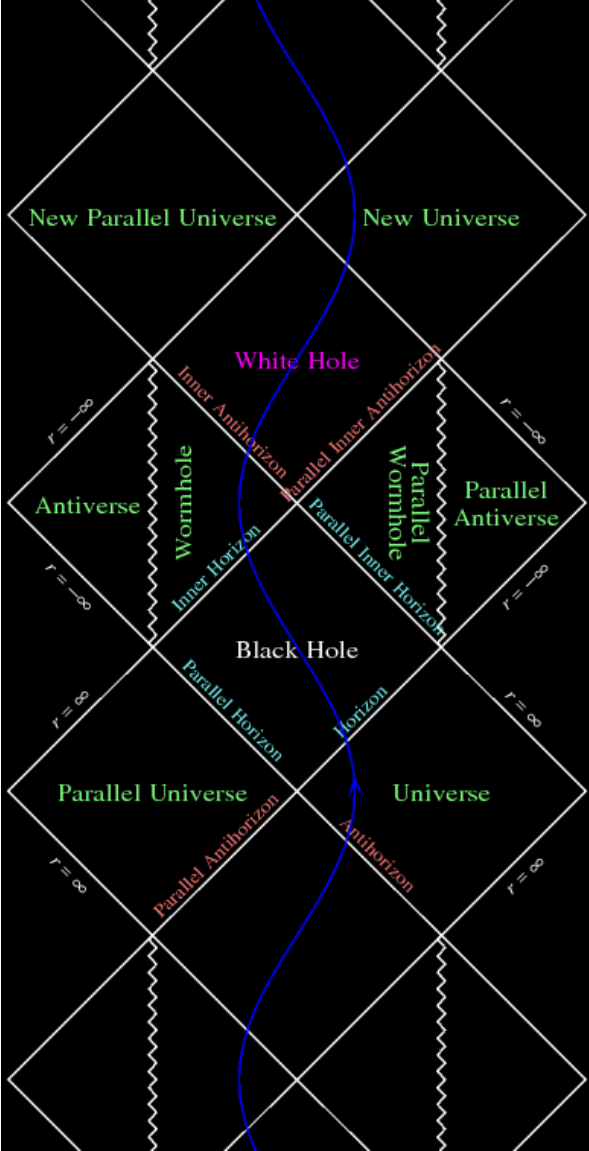
The mathematical solutions for super-extremal Reissner-Nordström black holes, those with charge exceeding mass, have no horizons at all, which is an even more drastic change in topology.

Real charged particles such as electrons and protons have charge greatly exceeding their mass. For example, an electron has a charge-to-mass of $e/m_e \approx 10^{21}$ (the ratio of the square root of the fine structure constant to the electron mass in Planck units). However, the character of such particles is dominated by quantum mechanics, not by classical general relativity.



Penrose diagram of a Kerr black hole

The Penrose diagram of the Kerr geometry for a rotating black hole looks like the Penrose diagram of the Reissner-Nordström geometry for a charged black hole, except that it is possible to pass through the disk bounded by the ring singularity of the rotating black hole to an antiverse.



References

1. Roger Penrose (1964) “Conformal treatment of infinity” in *Relativity, groups and topology* ed. C. de Witt & B. de Witt (Gordon and Breach, New York) p. 563-584; republished (2011) *Gen. Rel. Grav.* **43** 901-922 [10.1007/s10714-010-1110-5](https://doi.org/10.1007/s10714-010-1110-5)

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